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A scoping review of literature on climate change impacts among smallholder maize producers in low- and medium-income countries

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Climate change is redefining global agriculture through temperature swings, erratic rainfall, and frequent extreme events. According to the IPCC agricultural productivity growth in Sub-Saharan Africa has declined by approximately 5–6% due to climate warming, with maize yields particularly affected by temperatures exceeding 30–35 °C during critical growth stages. Smallholder farming communities face disproportionate vulnerability to intensifying climate hazards, yet knowledge regarding specific mechanisms through which climate variability affects smallholder maize producers—and evidence on adaptation strategy effectiveness—remains fragmented and methodologically limited. This is particularly concerning in Sub-Saharan Africa, where over 70% of maize producers experience high exposure and sensitivity alongside low adaptive capacity. This scoping review assessed gaps in empirical research on climate change impacts and adaptation responses among smallholder maize producers in LMICs. Following PRISMA guidelines, a systematic search identified 86 peer-reviewed articles published between 2008 and 2025 and indexed in Scopus, Web of Science, and AGRICOLA. After removing duplicates using Zotero and excluding qualitative-only studies, grey literature, studies with undefined outcome variables, and those without inferential statistics, 38 empirical peer reviewed articles employing quantitative or mixed method were analyzed using descriptive statistics in Microsoft Excel. The analysis revealed three main outcome variables: adaptation strategies (70.3%, $n = 26$), perception (21.6%, $n = 8$), and climate awareness (8.1%, $n = 3$). Geographically, studies concentrated heavily in Ethiopia (19%), Ghana (19%), and Zimbabwe (14%), while South Africa, Lesotho, Nigeria, Tanzania, Botswana, Burkina Faso, Cameroon, DRC, Eswatini, and Kenya each accounted for less than 10%, revealing clear research bias. Cross-sectional research design dominated ($n = 32$, 86%), followed by exploratory design ($n = 4$, 11%) and longitudinal design ($n = 1$, 3%). Regarding theoretical frameworks, 28% of studies lacked any framework, 20% employed random utility maximization theory, and 52% drew on 13 other theories including adaptive capability, expected utility, protection motivation, socioecological system, and vulnerability theory— each accounting for 4%. Notably, no study incorporated moderation variables. The review reveals that while adaptation behaviors are well-documented, empirical evidence quantifying actual climate impacts on productivity, income, and awareness remains scarce. Future research should prioritize impact studies evaluating scalable interventions and utilize robust resilience metrics to strengthen smallholder maize systems.

KEYWORDS

Climate adaptation strategies, climate change, Climate impact, maize producers, resilience, smallholder farmers, sustainability

1 Introduction and background

Maize is one of the world's most critical staple crops, sustaining food security and livelihoods for millions of smallholder households, especially in Sub-Saharan Africa (SSA), where production systems are predominantly rainfed and resource-constrained (Fan and Rue, 2020; Shiferaw et al., 2011). Since the 1960s, global maize area has nearly doubled to approximately 202 million hectares by 2020, while average yields have roughly tripled from 2 to 5.8 tonnes per hectare (Erenstein et al., 2022). Maize now contributes 25–50% of total calories in many developing regions, with over 30% of dietary calories in parts of Africa and Latin America derived from this crop (Lowder et al., 2021).

Despite growing recognition of climate threats to maize systems, the empirical literature remains skewed toward documenting adaptation choices rather than rigorously quantifying measured climate impacts and resilience outcomes (Mulungu and Ng'ombe, 2019; Mnukwa et al., 2025). This imbalance matters because smallholder maize producers operate under structural constraints—limited access to irrigation, credit, extension, and soil fertility inputs—that shape what is feasible to adopt, how costly it is to adapt, and whether adaptation actually translates into stable yields, food security, and income gains (Hassan and Nhemachena, 2008; Kori, 2023; Dimkpa et al., 2023).

Consequently, although climate risks to smallholder maize systems are widely acknowledged, the empirical literature still offers insufficient, uneven, and methodologically limited evidence on what climate change is doing to productivity and livelihoods, which adaptation strategies measurably work, and under what contexts and constraints they succeed or fail (Mnukwa et al., 2025; Muoni et al., 2025).

1.1 Global impact of climate change on smallholder maize farmers

Maize supports the food security and livelihoods of hundreds of millions of smallholder farmers, particularly in Sub-Saharan Africa, Latin America, and parts of Asia (Lowder et al., 2021; Ricciardi et al., 2018). Despite recent record harvests, maize systems face mounting threats as climates warm. Rising temperatures, shifting rainfall patterns, and more frequent extreme weather events—including heat waves, droughts, and floods—increasingly undermine maize productivity and stability for smallholders worldwide (IPCC, 2022a, 2022b; Wheeler and von Braun, 2013).

1.1.1 Climate threats to maize production

Maize production is increasingly exposed to a range of climate-related stresses that directly undermine yields and farmer resilience. These threats manifest through multiple pathways, from rising temperatures and shifting rainfall patterns to escalating pest pressures and soil degradation. The following key challenges highlight the most critical climate risks confronting smallholder maize systems worldwide:

- *Temperature stress* poses a critical challenge. Many maize-growing areas, especially in the tropics and subtropics, are experiencing

hotter growing seasons. Maize is particularly sensitive to high temperatures during pollination, with yield losses occurring when daytime highs exceed 30–35 °C (De Groote et al., 2020; Ranum et al., 2021). Climate models project that each 1 °C of warming will reduce tropical maize yields by approximately 5% on average (Karwani G. M. et al., 2024; Karwani M. et al., 2024; Nelson et al., 2022). For example, IPCC reports indicate that maize yields in West Africa are expected to decline by approximately 9% at 1.5 °C warming and by 41% at 4 °C relative to 2005 levels if no new adaptations occur (Kapari et al., 2023). Recent analyses project up to 40% declines in U. S. maize yields by late century under high-emissions scenarios, with similar losses projected in China and parts of southern Africa (Rodríguez-Barillas et al., 2024; Strandt, 2025).

- *Water stress and rainfall variability* present equally severe challenges. Approximately 70–95% of global maize is rainfed (Nelson et al., 2022), making changes in precipitation patterns critical. In many smallholder regions, rainfall is becoming more erratic and drought frequency is rising. Periods of late-season drought are particularly damaging to maize, reducing kernel number and grain fill, while intensifying monsoons and heavy rainfall can cause waterlogging or soil erosion (Tchouandem Nzali et al., 2024). Empirical studies confirm that agricultural productivity growth in Sub-Saharan Africa has already been slowed by climate change, with maize yield growth approximately 5–6% lower than it would have been without warming (IPCC, 2022a, 2022b).
- *Pests, diseases, and soil degradation* compound these challenges. Warming climates are expanding the range and seasonality of pests such as fall armyworm and stem borers, and pathogens like maize lethal necrosis. In Sub-Saharan Africa, pests and diseases already contribute 10–35% yield losses in maize (Lamichhane et al., 2022). Higher temperatures and humidity also accelerate soil degradation processes, reducing fertility on already marginal smallholder plots. Some models estimate up to a 50% increase in pest-related maize losses by mid-century (Lamichhane et al., 2022; Utobo and Asogwa, 2021).

1.1.2 Projected global impacts and smallholder vulnerability

Climate change modeling consistently projects net negative impacts on maize productivity globally. A comprehensive analysis accounting for real farmer adaptations estimates that by 2100, global maize yields will be 13–28% below what they would be without climate change, with a median decline of 28% under RCP8.5 (Manzvera et al., 2024). Mid-21st century projections show most of North America, China, southern Africa, and parts of Europe suffering yield declines often exceeding 20–40% (Strandt, 2025).

These climate risks interact disproportionately with longstanding smallholder constraints. Smallholder maize farmers often cultivate marginal rainfed lands with limited irrigation, use few inputs, and have minimal access to credit or high-yielding varieties. Consequently, their baseline yields and resilience are lower than those of commercial farms. While global average maize yields are approximately 5.8 tonnes

per hectare (De Groote et al., 2020), many African smallholder plots yield only 1–2 tonnes per hectare. The IPCC emphasizes that communities with climate-sensitive livelihoods are disproportionately vulnerable when climate hazards intensify, with vulnerability amplified by poverty and limited infrastructure (IPCC, 2022a, 2022b).

1.1.3 Global climate policy frameworks

Over the past decade, international climate policy has increasingly recognized agriculture as a critical adaptation sector. The Paris Agreement (2015) explicitly calls for enhancing adaptive capacity in vulnerable sectors (Article 7), and subsequent COP decisions have sought to integrate agriculture into global climate strategies. Major institutions now emphasize climate-smart agriculture (CSA) and resilience-building. The World Bank's Climate Change Action Plan (2021–2025) identifies agri-food systems as a key transition for meeting Paris targets and has scaled up climate-smart agriculture financing eightfold to approximately \$3 billion per year since Paris (World Bank, 2025). However, current climate finance is heavily skewed: agrifood systems cause approximately 30% of global emissions but receive only 4% of climate funds, and just 20% of that reaches smallholders (Food and Agriculture Organization of the United Nations, International Fund for Agricultural Development, UNICEF, World Food Programme, and World Health Organization, 2023).

1.2 Impact of climate change on smallholder maize farmers in Sub-Saharan Africa

Maize is the dominant staple crop across Sub-Saharan Africa, occupying approximately 41 million hectares—roughly 40% of the continent's cereal area—with production rising from 67 to 91 million tons between 2010 and 2020 (Erenstein et al., 2022). Despite this expansion, average SSA maize yields remain critically low at approximately 2 tonnes per hectare, representing only 30–50% of achievable potential and lagging significantly behind global yields. Smallholder farmers, who cultivate roughly 80% of SSA's maize on predominantly rainfed systems (95% of African farms lack irrigation), are particularly vulnerable to climate variability (Erenstein et al., 2022). Across Sub-Saharan Africa, 71% of smallholder households experience climate shocks, 54% endure at least one food-insecure month each year, and 80% lack the resources to adapt. Notably, over 70% of these maize producers face high exposure and sensitivity alongside low adaptive capacity, underscoring the urgent need for targeted, context-specific interventions (Ogunyiola et al., 2022).

1.2.1 Regional climate impacts

Climate change does not affect maize production uniformly across Sub-Saharan Africa; instead, its impacts vary widely by region, reflecting differences in agroecological conditions, rainfall patterns, and exposure to extreme events. Understanding these regional dynamics is essential, as they reveal both the diversity of challenges smallholder farmers face and the need for context-specific adaptation strategies. The following highlights illustrate how climate variability manifests across Southern, Western, and Eastern Africa:

- *Southern Africa* faces acute drought risk linked to El Niño climate oscillations. The 2023/24 El Niño event brought extreme dryness

to Zambia, Malawi, and Zimbabwe, with February–March 2024 ranking among the driest months on record in many maize-growing areas. South African maize yield projections fell approximately 17% below normal levels (Nelson et al., 2022). Climate models project that above 1.5 °C of warming, drought frequency and duration will rise across much of Southern Africa, with critical precipitation declines in core maize areas (African Union, 2022; Nelson et al., 2022).

- *Western Africa's* maize belt—encompassing Nigeria, Ghana, Burkina Faso, Mali, and surrounding countries—has experienced pronounced warming trends and erratic rainfall patterns. Average maize yields in West Africa remain extremely low at approximately 1.2–1.3 tonnes per hectare (Erenstein et al., 2022; Sadiq et al., 2019). Extended dry spells diminish planting windows, while episodes of waterlogging occur during unusually heavy rains. High temperatures, often exceeding 30–35 °C during the growing season, further stress maize at flowering. The rapid spread of fall armyworm and invasive grain borers produces large year-to-year swings in national output, with yield losses ranging from 20–40% in infested areas (Njeru et al., 2023; Ogunyiola et al., 2022).
- *Eastern Africa* exhibits diverse climates—from the Ethiopian highlands and Tanzanian midlands to Uganda's lake basin—all facing mounting climate stress. Countries such as Ethiopia and Tanzania have experienced rapid expansion of maize cultivation area yet yields remain well below potential; Ethiopia's maize yields average under 4.5 tonnes per hectare despite favorable highland conditions (Erenstein et al., 2022). The Horn of Africa endured its worst drought in 40 years in 2021, devastating harvests across Ethiopia, Somalia, and adjacent areas (Karwani G. M. et al., 2024; Karwani M. et al., 2024; Nelson et al., 2022). Eastern African smallholders face a convergence of drought risk, heat stress, and pest outbreaks that produce increasingly unpredictable harvests.

1.2.2 Climate adaptation strategies and policy frameworks

In response to these challenges, smallholder maize farmers employ diverse adaptation strategies. Common responses include drought-tolerant or early-maturing varieties and planting-date adjustments to match shortened rainy seasons (Atinkut and Mebrat, 2016; Bedeke et al., 2019). Farmers also diversify cropping systems through intercropping and agroforestry to spread risk and improve soil fertility (Giller et al., 2021; Gugissa et al., 2022). Soil and water conservation practices—minimum tillage, mulching, integrated nutrient management, and water harvesting—are used to conserve moisture and protect soil health (Dimkpa et al., 2023; Bedeke et al., 2018). Evidence shows that adaptation can affect performance outcomes such as technical efficiency and productivity, yet effects vary by setting and strategy (Adzawla and Alhassan, 2021; Kamuzora, 2023).

At the continental level, African Union policy frameworks explicitly recognize climate threats to maize systems. Agenda 2063 prioritizes climate-resilient economies (African Union, 2022), while the 2014 Malabo Declaration reaffirmed commitments to transform agriculture by doubling productivity through sustainable practices (African Union Commission, 2015a, 2015b). However, implementation has lagged, with many countries' National Agriculture Investment Plans initially failing to adequately address climate adaptation. The African Union's, 2022–2032 Climate Change and Resilient

Development Strategy recommend integrating climate adaptation measures within staple crop systems (African Union, 2022).

1.2.3 Evidence gaps and research priorities

A major limitation of the current evidence base is its uneven distribution. Maize-focused climate adaptation studies are concentrated in Ethiopia and Ghana, with fewer rigorous empirical studies in other highly vulnerable contexts (Bedeke et al., 2019; Aidoo et al., 2021). Evidence from semi-arid Zimbabwe emphasizes how climate vulnerability and structural drivers—inputs, soil constraints, and farming conditions—intersect to sustain low yields, reinforcing the need to analyze climate impacts together with underlying agronomic and institutional constraints (Muoni et al., 2025; Dimkpa et al., 2023). The field needs stronger evidence on the magnitude of climate impacts, the effectiveness of specific adaptation bundles, and the distribution of benefits and burdens across different types of smallholder households (Abate et al., 2017; Olagunju et al., 2020; Kariuki, 2018).

1.3 Impact of climate change on smallholder maize farmers in Kenya

Smallholder maize farming remains the backbone of Kenya's food security and rural economy, employing millions, with the agriculture sector contributing approximately 22.5% of GDP [Kenya National Bureau of Statistics (KNBS), 2025]. Despite its critical importance, the sector faces escalating climate-driven challenges that threaten productivity, incomes, and livelihoods. Over recent decades, Kenya has experienced increased frequency and severity of droughts, erratic rainfall patterns, and rising temperatures—factors that compound the vulnerability of maize-dependent smallholders (Kabara et al., 2022).

Evidence from major production areas, including Trans Nzoia, Uasin Gishu, and Bungoma, reveals yield declines of 15–25% over the past decade due to prolonged dry spells and unpredictable rainfall (Njeru et al., 2023). Temperature increases of just 1 °C above long-term averages have shortened maize growth periods, reducing grain yields by up to 20% (Njeru et al., 2023). The Rift Valley region, which contributes nearly half of national maize production, exemplifies how central smallholders are to the national food supply (StatsKenya, 2025).

Beyond climatic stressors, biological threats intensify production challenges. Warmer, more humid conditions have expanded the distribution of pests and diseases—particularly fall armyworm and maize streak virus—resulting in yield losses ranging from 20 to 40% in affected areas (De Groote et al., 2020; Ogunyiola et al., 2022). Simultaneously, soil erosion, nutrient depletion, and declining organic matter degrade already fragile smallholder plots, diminishing their capacity to withstand environmental shocks (Bryan et al., 2022).

Although national efforts promote drought-tolerant maize varieties, conservation agriculture, and agroforestry, adoption remains uneven and low—especially among women and resource-poor farmers—due to financial, informational, and institutional barriers (Atube et al., 2021). High costs for improved inputs, limited access to credit, under-resourced extension services, and insecure land tenure continue to constrain the use of climate-resilient technologies across Kenya's rural counties (Wanjira et al., 2022).

These constraints expose smallholders to recurrent yield losses and income instability, forcing many into cycles of debt, asset loss, and reduced farm investment. Despite considerable global and regional research on climate change impacts in agriculture, significant

knowledge gaps remain regarding the specific vulnerabilities of smallholder maize farmers in Kenya across diverse agroecological conditions and socioeconomic contexts. Existing research often aggregates findings across regions, masking the distinctly different vulnerabilities of smallholders in high-potential areas such as Trans Nzoia and Uasin Gishu compared to those in semi-arid counties like Machakos and Makueni. This lack of localized, context-specific evidence limits targeted climate adaptation planning and effective policy formulation.

Furthermore, most literature focuses on either specific climatic impacts or adaptation practices in isolation, without integrating the structural and institutional dimensions that shape farmers' responses. Evidence remains limited on how gender, credit access, and extension services influence adaptive decisions at the individual smallholder level. Consequently, existing policies and programs often adopt blanket approaches unable to capture the localized realities of smallholders facing different climatic and socioeconomic challenges. Addressing these gaps is essential to maintain Kenya's food security, safeguard rural livelihoods, and advance national development goals under the Kenya Climate Smart Agriculture Strategy 2022–2032 and Vision 2030.

Data from KNBS (2025) indicate that Kenyan smallholder households contribute approximately 75–80% of national maize output, with the majority cultivating holdings of less than two hectares and mean yields reported at 1.5–1.8 tonnes per hectare (De Groote et al., 2020; Njeru et al., 2023). Yet despite growing recognition of these challenges from global to regional scales, the pathway from climate vulnerability to resilience remains largely undefined for smallholder maize farmers—whether across Sub-Saharan Africa's diverse agroecological zones or within Kenya's 4.5 million farming households—a gap this research seeks to address.

2 Purpose of the scoping literature review

This scoping review aimed to systematically map and synthesize the empirical evidence base on climate change impacts and adaptation among smallholder maize producers in Low and Medium Income Countries (LMICs). Despite growing recognition of climate threats to smallholder maize systems, the existing literature remains fragmented across geographic contexts, outcome measures, and methodological approaches. This review sought to assess the extent, nature, and quality of empirical research in this domain, identify dominant themes and research gaps, and provide evidence-based recommendations for future research, policy, and practice. By clarifying what has been studied, where research is concentrated, which outcomes have received attention, and what critical evidence gaps persist, this review contributes to building a more coherent knowledge base for supporting climate-resilient smallholder maize production systems.

2.1 Study rationale and objectives

Climate change poses existential threats to smallholder maize producers in LMICs, yet the empirical evidence base documenting these impacts and evaluating adaptation responses remains methodologically limited, geographically uneven, and thematically skewed. While adaptation strategies have been extensively documented, far less attention has been paid to measuring actual climate impacts on

productivity and livelihoods, understanding farmers' perceptions and awareness of climate risks, or evaluating the effectiveness and scalability of adaptation interventions across diverse contexts. Furthermore, as is evidenced in this review, research is heavily concentrated in a few countries (notably Ethiopia, Ghana, and Zimbabwe), leaving other vulnerable regions underrepresented. Methodologically, the dominance of cross-sectional designs limits understanding of temporal dynamics, while the absence of mediating and moderating variables obscures the pathways and conditions through which adaptation occurs.

This scoping review addresses these limitations by systematically examining peer-reviewed empirical studies published between 2008 and 2025. The review identifies dominant research themes, assesses the geographic and methodological distribution of evidence, evaluates theoretical frameworks employed, and highlights critical gaps in knowledge. By synthesizing findings across adaptation strategies, farmer perceptions, and climate awareness, this study provides a comprehensive assessment of the current state of knowledge and charts directions for advancing both scholarship and practice in support of climate-resilient smallholder maize systems.

The general Objective of the scoping review was to map and synthesize the empirical evidence on climate change impacts, adaptation strategies, and farmer responses among smallholder maize producers in Sub-Saharan Africa, while identifying key knowledge gaps that constrain the development of effective, equitable, and sustainable pathways to resilience.

The specific objectives were as follows:

1. To examine the scope, geographic distribution, methodological approaches, and theoretical frameworks used in empirical studies on climate change impacts and adaptation among smallholder maize producers in low- and middle-income countries between 2008 and 2025.
2. To assess how smallholder maize farmers perceive and understand climate change risks, and how these perceptions influence their adaptive behaviors and responses.
3. To document and synthesize the adaptation strategies empirically reported by smallholder maize producers, including the determinants of their adoption, any evidence of their effectiveness, and how farmer perceptions influence responses.
4. To critically analyze methodological, theoretical, geographic, and contextual gaps in the existing empirical literature, and evaluate their implications for advancing knowledge on effective, equitable, and sustainable climate resilience in smallholder maize systems

2.2 Overarching research question

What does empirical research reveal about climate change impacts, adaptation strategies, and responses among smallholder maize producers in Low- and Medium-Income Countries, and which gaps constrain effective and equitable resilience pathways?

2.2.1 Specific research questions

1. What is the scope and distribution of empirical research on impact of climate change and smallholder maize production in LMICs between 2008 and 2025?

2. How do smallholder maize farmers perceive and understand climate change, and how do these perceptions influence adaptive behavior?
3. What adaptation strategies have been empirically documented among smallholder maize producers, and what evidence exists regarding their effectiveness and adoption determinants?
4. What methodological, theoretical, geographic, and contextual gaps exist in empirical literature, and what are their implications for research, policy, and practice?

3 Methodology

This scoping review mapped and synthesized empirical evidence on climate change adaptation among smallholder maize farmers in Low- and Medium-Income Countries. A scoping review design was selected because it facilitates comprehensive examination of the nature, scope, and breadth of research activities within multidisciplinary fields, accommodating diverse methodologies, theoretical perspectives, and outcome measures (Arksey and O'Malley, 2005; Levac et al., 2010). This approach was particularly appropriate given that literature on smallholder maize adaptation spans agricultural economics, environmental sciences, and rural sociology.

The review followed the PRISMA Extension for Scoping Reviews (PRISMA-ScR) framework (Tricco et al., 2018; Page et al., 2021), which provides structured guidance through clearly defined stages of identification, screening, eligibility, and inclusion, enhancing methodological rigor, transparency, and reproducibility. A scoping search was conducted for peer-reviewed empirical studies published between 2008 and 2025, capturing climate adaptation trends following key policy milestones including the 2015 Paris Agreement and Africa's 2022 Malabo Declaration on climate-resilient agriculture (African Union Commission, 2015a, 2015b).

3.1 Search strategy

A comprehensive literature search across multiple databases yielded a total of 120 records before deduplication. A comprehensive literature search across multiple databases yielded a total of 120 records before deduplication. The number of papers retrieved from each source database included Scopus ($n = 48$), Web of Science ($n = 38$), Google Scholar ($n = 17$), ScienceDirect ($n = 9$), SpringerLink ($n = 4$), and AGRICOLA ($n = 4$). After merging results and removing duplicates using Zotero, the search yielded 86 unique studies for screening. This rigorous identification process, following PRISMA-ScR guidelines, ensured broad coverage while avoiding double-counting. The above breakdown of sources can be reported in the Methodology (Search Strategy) section or illustrated in a PRISMA flow diagram under "Identification" to show contributions of each database.

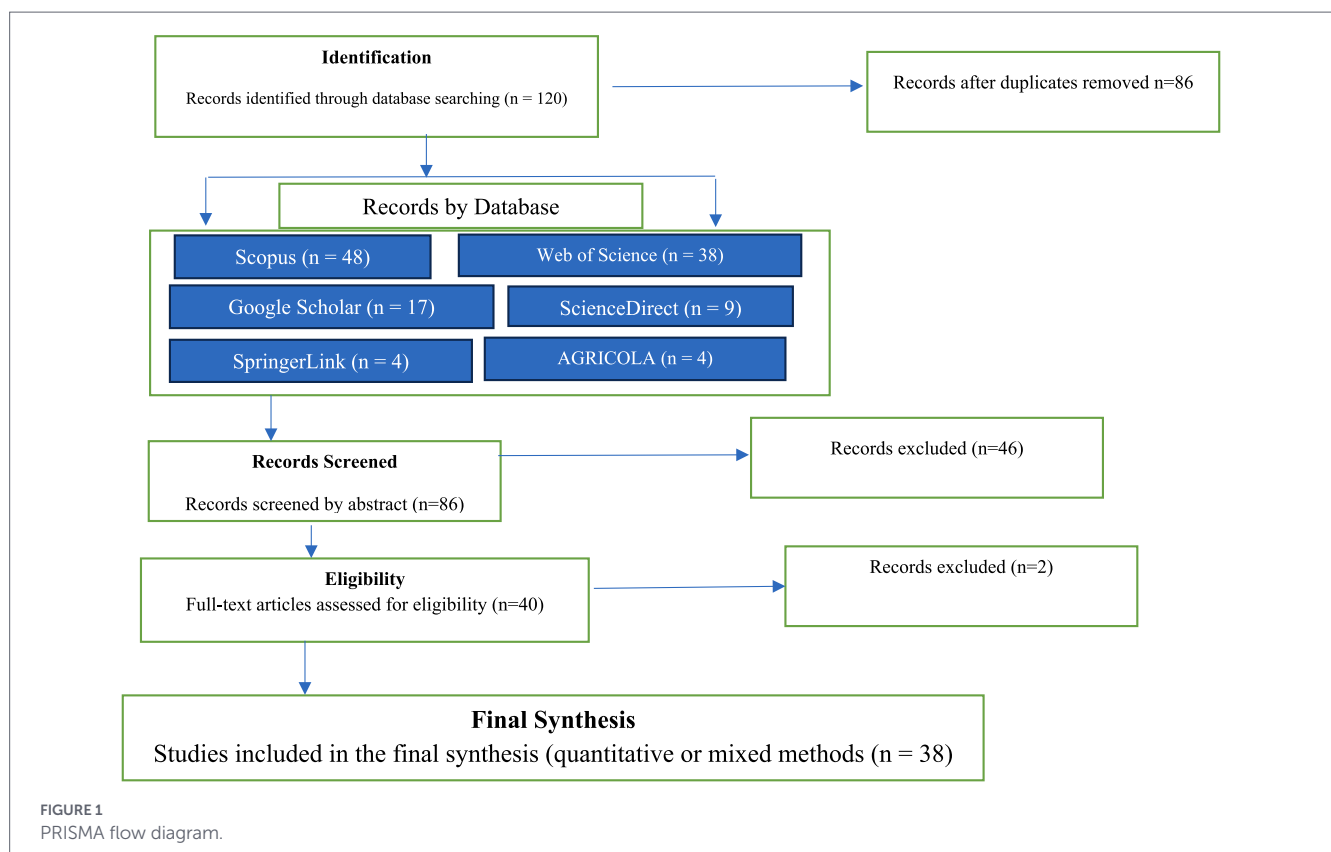
Following title/abstract screening and full-text eligibility checks, 38 studies were ultimately included in the scoping review. Notably, the vast majority of these 38 articles were found via mainstream indexing databases (Scopus and Web of Science). In fact, over 30 of the included studies were originally identified through Scopus, and a very high overlap with Web of Science (≈ 25 –30 studies) was observed, reflecting the comprehensive indexing of relevant journals in these two databases. By contrast, only a handful of final studies came from the other

sources – for example, a few unique inclusions were captured through Google Scholar and AGRICOLA (e.g., an extension research article not indexed elsewhere), while ScienceDirect and SpringerLink searches did not yield any additional unique studies beyond those found in Scopus/WoS. In sum, Scopus and WoS accounted for the lion's share of included articles, underscoring their importance, whereas specialized or publisher-specific databases contributed relatively few unique studies. These source-attribution insights can be briefly noted in the Methodology section (e.g., “Most included studies were identified via Scopus/WoS, with few unique records from Google Scholar or AGRICOLA”), reinforcing the search strategy's effectiveness. Focusing on the studies obtained from Scopus and Web of Science (which overlap with a considerable number of articles included), several notable patterns in terms of study context and content:

- **Countries of study:** the Scopus/WoS-sourced studies spanned a range of low- and middle-income countries, with a heavy focus on Sub-Saharan Africa. Consistent with the overall dataset, Ethiopia, Ghana, and Zimbabwe were among the most frequent study locations (each appearing in multiple Scopus/WoS-sourced papers). This reflects a concentration of research on climate adaptation in these countries. Other countries like South Africa, Nigeria, Lesotho, Kenya, and Tanzania also featured, though in fewer studies (see Figure 1 below). This geographic emphasis aligns with the broader trend of climate-change impact studies clustering in certain African nations with high maize-dependence.
- **Common keywords/topics:** across the Scopus/WoS studies, certain themes and variables recurred frequently. Many papers examined

aspects of “climate change adaptation strategies” (e.g., adoption of drought-tolerant maize, soil and water conservation, livelihood diversification) as the primary outcome. Correspondingly, common independent variables and descriptors included terms related to farmers' perceptions of climate risk, awareness of climate information, and adaptive capacity. Socio-economic factors were also prominent: for instance, household size, education level, gender of the household head, access to credit, extension services, farm size, and tenure security were regularly analyzed as determinants of adaptation or food security outcomes. These recurring keywords suggest a core set of concerns across studies – linking climate variability with farmer decision-making and livelihood factors.

- **Recurring journals and authorship patterns:** the included studies from Scopus/WoS tend to cluster in a few key journals. Notably, the open-access journals Sustainability and Heliyon were common publication outlets, together accounting for nine of the 38 studies (5 in Sustainability, 4 in Heliyon). Other studies appeared in specialized journals like African Journal of Agricultural and Resource Economics, Agriculture & Food Security, and Journal of Human Ecology (each hosting multiple included papers). This indicates that research on smallholder climate adaptation is often published in sustainability and agriculture-focused journals with broad accessibility. In terms of authorship, the dataset did not reveal a single dominant author or research group – instead, a diverse range of scholars (often based in or collaborating with researchers in Africa) authored the studies. A few authors did appear in more than one included article (suggesting active researchers in this niche), but generally authorship was distributed, reflecting contributions from various independent studies across countries. These



content insights can enrich the Results section (perhaps under a subheading like “Profile of Included Studies”), highlighting prevalent study themes and publication venues.

A comprehensive search strategy was employed to ensure inclusivity and reduce selection bias. Relevant peer-reviewed literature was retrieved from five reputable databases: Scopus, Web of Science, Science Direct, Springer Link, and Google Scholar. A standardized Boolean search string was applied uniformly across all platforms: (“climate change” OR “climatic variability”) AND (“smallholder” OR “small-scale farmer”) AND (“maize” OR “corn”) AND (“adaptation” OR “resilience”). Database-specific search strategies were employed to account for variations in indexing, retrieval algorithms, and syntax, as highlighted in comparative studies of bibliographic databases (Zhu and Liu, 2020, for Scopus vs. WoS; Gusenbauer and Haddaway, 2020, for unique features on Google Scholar). While core search terms remained consistent across platforms to ensure conceptual alignment, queries were tailored to the capabilities of each database. The use of these databases ensured the search covered potential gaps in coverage for interdisciplinary or emerging topics without eligibility or access constraints tied to institutional registration and publisher agreements, which can limit broad, transparent retrieval of non-subscription content.

This standardized approach maintained consistent inclusion parameters across all databases: peer-reviewed articles published between 2008 and 2025, written in English, and focusing on smallholder maize production in low- and middle-income countries with emphasis on Sub-Saharan Africa. Searches were restricted to English-language publications for consistency, as English is the dominant language of scholarly communication in agricultural sciences (Peters et al., 2020). The same exclusion criteria were applied uniformly—excluding qualitative-only studies, reviews, commentaries, studies with ambiguous outcome measures, and those lacking inferential statistics. This approach minimized database-specific bias and ensured comprehensive capture of relevant empirical evidence across indexing platforms, enhancing reproducibility and transparency as recommended by PRISMA-ScR guidelines (Munn et al., 2022).

Zotero reference management software was used to identify and remove duplicates, yielding 86 unique studies that underwent title and abstract screening for eligibility. The rationale for this systematic approach lies in the necessity for transparency and reproducibility when undertaking literature mapping, with adherence to PRISMA-ScR ensuring consideration of all potentially relevant studies while avoiding subjective bias in article inclusion (Pham et al., 2014; Peters et al., 2015).

3.2 Document selection and eligibility criteria

After removing duplicate records, 86 studies were screened for relevance using predefined inclusion and exclusion criteria. The screening process was conducted by two independent reviewers to minimize bias and enhance reliability. Any discrepancies that emerged during the selection process were discussed and resolved through consensus to ensure objectivity and methodological rigor. Following this process, 38 empirical studies focusing on smallholder maize production and climate adaptation were retained for detailed review and synthesis.

The justification for these selection criteria lies in the need to ensure empirical robustness and analytical comparability across included studies. By prioritizing quantitative and mixed-methods designs, the review captured statistically measurable relationships between climate variables, adaptation strategies, and productivity outcomes—critical dimensions for developing actionable insights and informing policy recommendations (Bozzola et al., 2021) (Table 1).

3.3 PRISMA flow diagram

The PRISMA Flow Diagram illustrates the study selection process, showing the number of records identified, screened, excluded, and retained at each stage. This visual representation enhances transparency and aligns with PRISMA-ScR recommendations for documenting evidence synthesis (Page et al., 2021).

3.4 Data extraction and coding

To ensure methodological rigor, a comprehensive codebook was developed to standardize data extraction across included studies. This encompassed key domains including: dependent-variable themes such as climate change adaptation strategies, perceptions of climate risk, and awareness of climate impacts; independent-variable categories encompassing climatic factors (e.g., rainfall variability and temperature shifts), farm characteristics (e.g., landholding size and input intensity), socioeconomic factors (e.g., household income and education level), and institutional support mechanisms (e.g., extension services and credit access); methodological items such as research questions/hypotheses, mediation and moderation tests, theoretical frameworks, research design, sampling techniques, data-collection tools, and reliability/validity checks; and additional metadata including country/county context, target population, inferential analyses, key findings, limitations, assumptions, and calls for further research. A pilot coding phase was implemented to verify the codebook's clarity and consistency, involving random selection of 15% of the included studies for initial coding (Saldaña, 2021). Two independent coders applied the codebook, with discrepancies resolved through iterative refinement of definitions until inter-coder reliability thresholds were achieved (Table 2).

Thematic synthesis followed, wherein patterns within each dependent-variable theme were identified via thematic analysis. Coded entries were aggregated into thematic matrices organized by studies and subthemes, enabling calculation of subtheme frequencies and their co-occurrence with independent variables. This culminated in generation of thematic maps to visualize interconnections among adaptation strategies, perceptions, awareness, and food-security outcomes. Studies were further classified by methodological design (quantitative, qualitative, or mixed methods) and geographic context to support subgroup analyses. Contextual metadata, including country, county, target population characteristics, and sampling techniques, were recorded, with summary tables compiled to display study counts and salient methodological features (Pham et al., 2014; Peters et al., 2015).

3.5 Data analysis and thematic synthesis

A thematic synthesis approach was employed to aggregate evidence across studies and identify dominant patterns, in line with guidance by Braun and Clarke (2006) and Thomas and Harden (2008).

TABLE 1 Inclusion and exclusion criteria.

Criteria type	Category	Description	Number of studies
Number of original studies explored		Original Number of studies explored	120
Inclusion	Publication period	Published between 2008 and 2025	
	Publication type	Peer-reviewed journal articles	
	Study design	Quantitative or mixed-method research designs	
	Outcome measures	Explicitly defined dependent and independent variables	
	Research focus	Directly addressed smallholder maize production in the context of climate change or climate variability	
	Geographic scope	Low- and middle-income countries (LMICs), with emphasis on Sub-Saharan Africa	
Exclusion	Qualitative-only or theoretical designs	Studies employing only qualitative methods or proposing theoretical frameworks without empirical data	18
	Reviews and conceptual models	Literature reviews, commentaries, opinion pieces, and conceptual analyses without primary evidence	14
	Ambiguous outcome measures	Papers lacking measurable dependent variables or precise outcome definitions	8
	Methodological issues	Insufficient sample size, non-peer-reviewed outlets, or mixed methods without quantitative component	6

TABLE 2 Summary of research gaps in climate change adaptation literature for smallholder maize producers.

Gap category	Key gaps identified	Implications for research and policy
Theoretical gaps	Many studies lack explicit theoretical frameworks. Vulnerability theory dominates, while behavioral, socio-ecological, climate justice, and political economy perspectives are rarely integrated. Constructs such as perception and awareness are often treated as static and not grounded in psychological or social theories. No studies apply moderation frameworks.	Limits conceptual coherence and weakens explanation of why farmers respond differently to climate risks. Policies risk oversimplifying adaptation drivers and ignoring social, cultural, and equity dimensions.
Empirical gaps	Heavy reliance on cross-sectional, self-reported data. Limited use of objective climate or yield data. Few studies assess long-term adaptation effectiveness or causal impacts. Geographic concentration in Ethiopia, Ghana, and Zimbabwe leaves many vulnerable regions underrepresented.	Weakens causal inference and generalizability. Policymakers lack reliable evidence on which adaptation strategies deliver sustained benefits across contexts.
Contextual gaps	Insufficient attention to gender, ethnicity, land tenure, institutional trust, and historical marginalization. Indigenous knowledge systems and communication dynamics are rarely examined. Little analysis of how information is framed or trusted.	Interventions may be culturally misaligned, inequitable, or ineffective. Local realities and social power structures remain poorly understood.
Methodological gaps	Overuse of regression-based surveys with limited qualitative or participatory methods. Minimal use of longitudinal, experimental, GIS, or social network approaches. Measurement tools for perception and awareness lack validation.	Constrains understanding of dynamic adaptation processes and social learning. Limits policy-relevant insights into how adaptation evolves over time.
Analytical gaps (mediation and moderation)	No studies formally test mediating or moderating variables such as gender, education, extension access, or agro-ecological context. Perception and awareness rarely modeled as behavioral pathways.	Masks heterogeneity in vulnerability and adaptation capacity. Encourages one-size-fits-all policy responses rather than targeted, equity-sensitive interventions.
Outcome gaps	Focus is skewed toward adaptation strategies, with limited attention to productivity impacts, income, nutrition, and resilience outcomes. Awareness studies are rare and geographically narrow.	Prevents full assessment of whether adaptation improves livelihoods and food security, limiting evidence-based investment decisions.

n = number of studies; percentages based on total sample of 38 peer-reviewed studies published 2008–2025. Representative studies are illustrative examples, not exhaustive lists.

Quantitative results were summarized through frequency counts and cross-tabulations of adaptation strategies, while qualitative dimensions were analyzed using inductive coding. Thematic matrices were constructed to visualize relationships among adaptation practices, perceptions, and food-security outcomes.

This analytical choice was justified because thematic synthesis allows integration of diverse empirical findings—bridging quantitative and qualitative insights—to develop a holistic understanding of smallholder adaptation. It also facilitates mapping of causal linkages from climatic drivers to livelihood outcomes, which aligns with the review's objective of constructing a conceptual framework for smallholder maize adaptation (Popay et al., 2006; Dixon-Woods et al., 2005).

3.6 Conceptual mapping and integration

To integrate insights holistically, a conceptual framework (Figure 2) was constructed by mapping causal pathways from climatic variables to perceptions, adaptation strategies, and food-security outcomes. This involved annotating mediating variables (e.g., income diversification) and moderating factors (e.g., extension access), with iterative refinement against qualitative subthemes and theoretical constructs until conceptual convergence was attained (Tricco et al., 2018; Page et al., 2021).

The framework traces causal pathways linking climate signals such as drought frequency, rainfall variability, and temperature rise to farmers' risk perceptions, adaptation decisions, and livelihood outcomes. It integrates both direct and indirect relationships, recognizing that impacts of climate variability are mediated by socioeconomic and institutional factors, including income diversification, access to credit, extension services, and education. These mediating and moderating factors help explain heterogeneity in adaptive responses among smallholder maize farmers across different regions of Kenya and Sub-Saharan Africa. This conceptual integration aligns with the Sustainable Livelihoods Framework and Climate Resilience Theory, which together emphasize dynamic interactions between environmental shocks, household assets, and institutional support systems in shaping adaptation capacity (Scoones, 1998; Béné et al., 2012; Kabara et al., 2022; Karwani G. M. et al., 2024; Karwani M. et al., 2024b).

By connecting empirical findings to these theoretical underpinnings, the framework not only clarifies how smallholders respond to

climatic stressors but also identifies leverage points where policy interventions can strengthen resilience. The synthesis further highlights methodological rigor, evidence patterns, and persisting knowledge gaps across studies, providing a foundation for future research and informing adaptive policy design. For transparency and reproducibility, the manuscript incorporates comprehensive documentation, including the PRISMA flow diagram for study selection, aggregated study limitations, and author-recommended avenues for future research.

4 Findings

4.1 Overview

This scoping review synthesizes 40 peer-reviewed quantitative and mixed-methods studies on climate change impacts on smallholder maize farmers, revealing a complex landscape of adaptation strategies. The predominance of practices like drought-tolerant maize and agroforestry highlights efforts to enhance food security and resilience amid climatic variability. However, uneven exploration of socioeconomic outcomes and resilience mechanisms points to critical research gaps. Grounded in cross-sectional household surveys and advanced statistical methods, such as double-hurdle models and structural equation modeling, these findings illuminate the interplay of adaptation, farmer perceptions, and food security. However, they underscore the urgent need for research and policy to advance context-specific, scalable solutions for sustainable and climate-resilient maize farming systems.

4.1.1 Scope of research outcomes investigated

The scoping review identified three thematic areas addressed across the 38 selected studies: adaptation strategies (70.3%, $n = 26$), perception (21.6%, $n = 8$) and awareness (8.1%, $n = 3$). Adaptation strategies emerged as the most frequently examined dependent variable, underscoring their central role in climate resilience within maize-based farming systems. Perception was the second most studied outcome, with research highlighting how farmers' views—particularly of rainfall and temperature variability—strongly shape adaptation behavior. Awareness ranked third, with a smaller body of studies showing its influence on decision-making, risk management, and the adoption of sustainable practices among smallholder maize farmers (Figure 3).

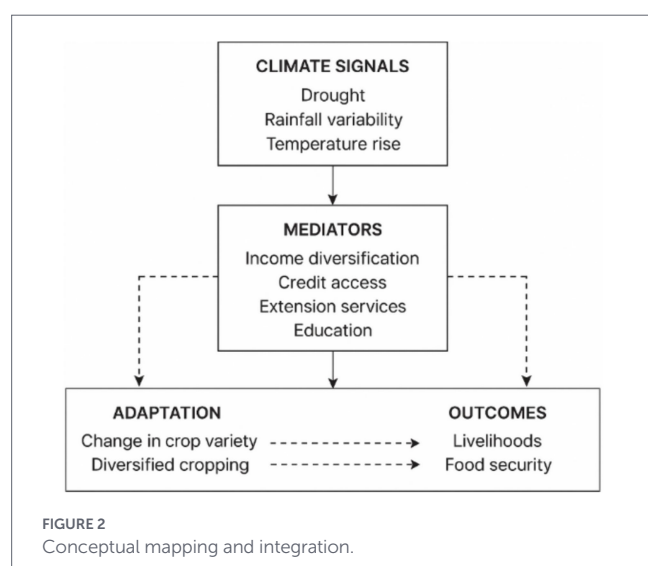
4.1.1.1 Geographic distribution of included studies

The geographic distribution of the 38 final included studies by country shows that the evidence base was highly concentrated in a few countries, with Ghana ($n = 7$) and Ethiopia ($n = 7$) contributing the highest number of studies, followed closely by Zimbabwe ($n = 7$). Together, these three countries account for more than half of all included studies, indicating a strong regional focus on climate change research among smallholder maize producers in these contexts.

South Africa contributed three studies, while Lesotho and Nigeria each accounted for two studies. Kenya, India, Eswatini, and the Democratic Republic of Congo (DRC) were each represented by one study.

Other countries such as Cameroon, Burkina Faso, Costa Rica, and Tanzania also appeared with one or two studies.

This uneven distribution highlights a strong concentration of research in Eastern and Southern Africa, with comparatively limited



representation from West Africa, Central Africa, and non-African regions. The pattern suggests that while climate adaptation among smallholder maize farmers has been widely studied in certain hotspots, many vulnerable regions remain underrepresented in the empirical literature (Figure 4).

4.1.1.2 Thematic focus of studies (adaptation, perception, awareness)

The distribution of thematic focus across countries shows the number of studies addressing adaptation strategies, climate change awareness, and farmer perception. Across nearly all countries, adaptation strategies dominated literature. For example, Ethiopia recorded five adaptation-focused studies, along with two perception studies, but no awareness-focused studies. Zimbabwe also showed a strong emphasis on adaptation (five studies), alongside two perception studies. Ghana exhibited the most balanced thematic coverage, with three adaptation studies, three perception studies, and one awareness study, making it the only country with representation across all three thematic areas.

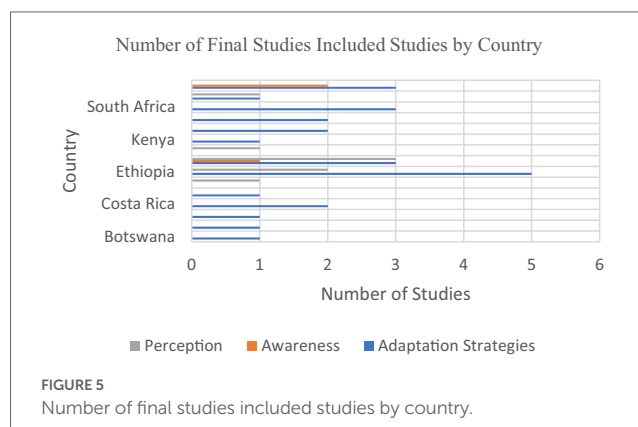
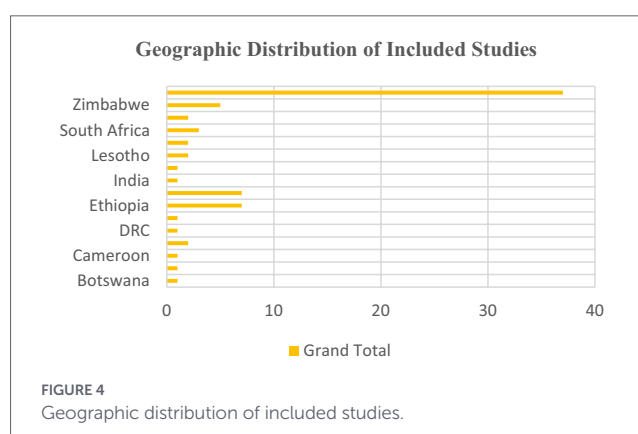
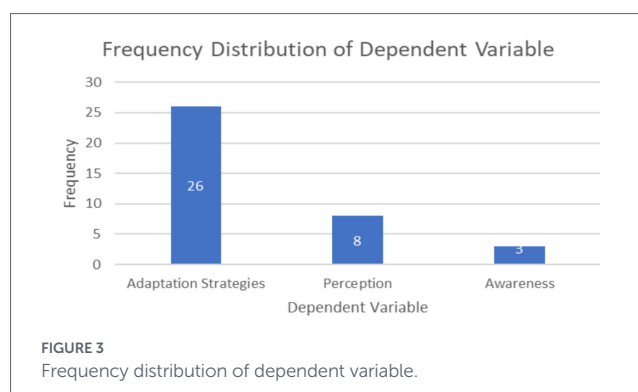
In contrast, Eswatini had only one awareness-focused study, with no studies on adaptation or perception. India was represented by a single perception-focused study, while Kenya, Botswana, Burkina Faso, Cameroon, DRC, Lesotho, Nigeria, South Africa, and Tanzania were represented exclusively by adaptation-focused studies (Figure 5). Overall, the results confirm that research on smallholder maize farmers has overwhelmingly prioritized adaptation strategies, while awareness and perception remain comparatively underexplored. Awareness studies were particularly rare and geographically limited, suggesting a major gap in understanding how knowledge and information influence adaptive behavior across different contexts.

The studies span diverse agro-ecological contexts across Sub-Saharan Africa (such as Nigeria, Ghana, Ethiopia, Zimbabwe, Kenya, Botswana, Cameroon, Lesotho, South Africa) and Central America (Costa Rica, Guatemala, Honduras), underscoring the global urgency of climate adaptation.

Mixed method approaches integrated structured questionnaires with focus group discussions, key informant interviews, and participatory ranking tools, providing valuable insights into social, institutional, and behavioral aspects of adaptation. Several studies employed theoretical frameworks, including the Random Utility Maximization Theory, Protection Motivation Theory (PMT), Expected Utility Theory, Socio-Ecological Systems, and Theory of Production. Sampling designs were rigorous, with many studies using agro-ecological zoning, climate vulnerability, and maize production as key inclusion criteria. Variables frequently investigated include gender, education, household size, access to credit and extension services, market proximity, land ownership, and climate information access.

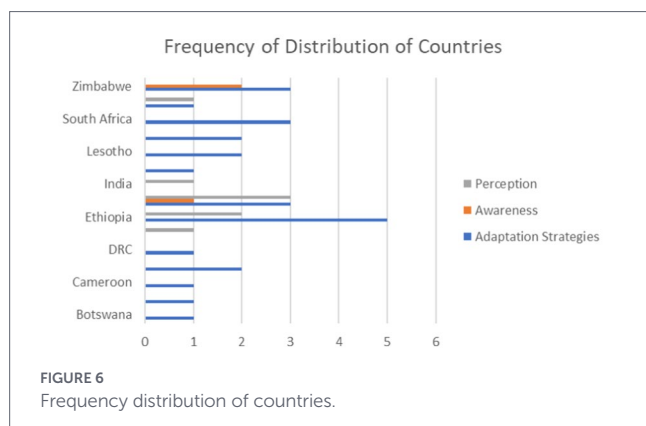
4.1.2 Countries of research

Studies on climate impacts among smallholder maize producers in LMICs are highly uneven, concentrated in Ethiopia (19%, $n = 5$), Ghana, South Africa, and Zimbabwe (each 12%, $n = 3$), followed by Costa Rica, Lesotho, and Nigeria (each 8%, $n = 2$). Botswana, Burkina Faso, Cameroon, DRC, Kenya, and Tanzania account for 4% each ($n = 1$). This geographic skew enhances context-specific insights for Eastern and Southern Africa but limits generalizability across Latin America, Southeast Asia, the Sahel, and Francophone Africa.



Adaptation strategy research spans 15 LMICs, with Eswatini leading at $n = 5$ studies and Ethiopia, Lesotho, and South Africa each contributing $n = 3$. India and Kenya record $n = 2$ studies, Costa Rica $n = 2$, and single studies in Botswana, Burkina Faso, Cameroon, DRC, Ghana, and Nigeria.

Perception-focused studies emphasize West and East Africa; Ghana's three studies across diverse districts illustrate agro-ecological and administrative heterogeneity, while one study each in Eswatini, India, and Tanzania indicates emerging attention in Southern Africa and South Asia (Figure 6). Awareness research is dominated by Zimbabwe, with multiple studies clustered in Manicaland Province, and a single study in Ghana's Northern Region. These imbalances risk skewing regional evidence on farmer vulnerabilities and adaptation uptake.



Underrepresented countries like Botswana, DRC, and Tanzania face acute climate threats yet contribute only one study each, underscoring critical evidence gaps. Strengthening geographic representation through participatory, multi-location, and longitudinal research will enrich understanding of local vulnerability profiles and adaptation capacity. Expanding coverage across additional LMICs and agro-ecological zones is essential to inform equitable, evidence-based policies that bolster resilience of smallholder maize systems globally and truly equitably.

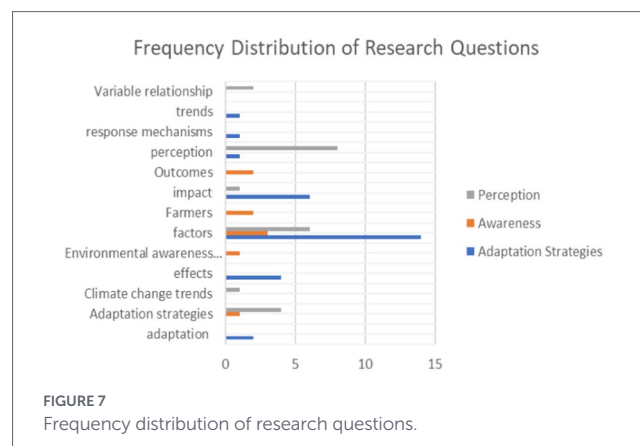
4.1.3 Research questions

Within 38 peer-reviewed studies of climate change impacts on smallholder maize producers, research questions concentrated on three main themes: perception, adaptation strategies, and environmental awareness. Perception-oriented inquiries dominated at 36% ($n = 8$), exploring how farmers recognize and interpret shifts in rainfall, temperature, and seasonality. These subjective experiences are deemed foundational to adaptation decision making. Questions on factors influencing adaptation featured in 27% ($n = 6$) of studies, examining socioeconomic, institutional, and resource-access determinants of practices like drought-tolerant varieties, soil and water conservation, and livelihood diversification.

Adaptation strategies as a direct focus comprised 18% ($n = 4$), yielding practical insights for scaling field-tested responses. Less attention was devoted to empirical relationships and biophysical diagnosis: 9% ($n = 2$) of questions addressed causal links between climate variables and yields or food security, while studies of long-term climate trends and direct impacts each accounted for 5% ($n = 1$).

In the broader literature, 48% ($n = 14$) of studies prioritized factors shaping adaptation, 21% ($n = 6$) assessed climate variability impacts on production, and 14% ($n = 4$) evaluated adaptation outcomes in resilience, food security, or income.

Cost considerations appeared in only 7% ($n = 2$), and environmental awareness and concern emerged in just 3% ($n = 1$), highlighting limited focus on farmers' knowledge and beliefs about climate risk (Figure 7). This distribution signals a behavioral and systems-oriented field that emphasizes individual perceptions and decision drivers over rigorous integration of biophysical evidence and awareness-building. Bridging this methodological gap will strengthen contextually relevant adaptation pathways. Expanding longitudinal, comparative studies and investing in climate literacy can deepen understanding of effective strategies and enhance sustainable farmer-led innovations.



4.1.4 Theories

Of the 38 studies reviewed, 26 employed theoretical frameworks, revealing substantial fragmentation in how climate change impacts among smallholder maize producers in low- and middle-income countries are conceptualized. Just under half (46%, $n = 12$) applied formal theory, while the majority (54%, $n = 14$) lacked explicit conceptual framing. Across theory-informed studies, the focus was primarily on farmer decision-making processes, vulnerability assessment, and adaptive capacity. Vulnerability theory was most frequently used ($n = 10$), particularly within adaptation strategies ($n = 8$) and perception analyses ($n = 2$), emphasizing sources of susceptibility to climate impacts and differential exposure shaped by socio-economic and environmental factors (Cáceres et al., 2024; Dilling et al., 2015; Ludena et al., 2015). The following points outline the specific theories and their areas of emphasis:

- *Economic rationality and utility maximization* approaches were evident but surprisingly limited. Random utility maximization (RUM), which focuses on farmers' choice behavior under uncertainty and assumes rational decision-making to maximize expected benefits (Carleton et al., 2024; Stetter and Mennig, 2024), appeared only once ($n = 1$), contrary to assumptions of its dominance in agricultural economics literature. The random utility framework has proven versatile in modeling adaptation choices under climate change, incorporating both deterministic and stochastic components to explain how individuals choose among alternatives in the presence of uncertainty (Baulcomb et al., 2022; Tabbo et al., 2018).
- *Behavioral and psychological frameworks* received minimal attention. Protection motivation theory ($n = 1$), which examines how threat appraisal and coping appraisal influence protective behaviors, appeared in one awareness study. Expected utility theory ($n = 1$), focusing on decision-making under risk (Holden and Tilahun, 2024; Piracci et al., 2023), and socio-cognitive theory ($n = 1$), emphasizing learning and social influences on behavior (Truelove et al., 2015), each featured once. Recent advances in behavioral decision theory, particularly Rank-Dependent Utility Theory, have provided richer understanding of farmer behavior under extreme environmental stress, revealing that risk preferences shift significantly in high-stakes climate situations (Holden and Tilahun, 2024).
- *Systems-oriented approaches* were underutilized. Socioecological systems (SES) theory ($n = 1$), which examines interactions

between social and ecological components of adaptive systems, recognizing them as complex and integrated structures shaped by continuously evolving dynamics including feedbacks, non-linearity, and cross-scale interactions (Giupponi et al., 2022; Hossain et al., 2024; Preiser et al., 2018), and adaptive capability theory ($n = 1$), focusing on dynamic capacity to adjust to changing conditions, each appeared once, despite their relevance to understanding complex farmer-environment interactions. Understanding and managing these feedbacks, non-linearity, and delays are key principles for enhancing both short- and long-term resilience in agricultural systems (Fedele et al., 2019; Galappaththi et al., 2019).

- *Production and resource management theories* showed limited application. Neoclassical production theory ($n = 1$), examining input-output relationships and efficiency, sustainable intensification theory ($n = 1$), focusing on productivity enhancement with environmental conservation, and Marshallian demand theory ($n = 1$), analyzing market responses to price changes, each appeared once.
- *Cultural and philosophical perspectives* were marginally represented. Cultural theory ($n = 1$), examining how worldviews shape risk perception, deductive theory ($n = 1$), and existence theory ($n = 1$) each featured once, highlighting minimal engagement with socio-cultural dimensions of adaptation.

Among theory-informed studies, vulnerability theory dominated ($n = 10$), particularly within adaptation strategies ($n = 8$) and perception analyses ($n = 2$). The conceptualization of vulnerability has evolved substantially, with scholars recognizing that vulnerability in climate change contexts encompasses multiple interacting factors and that different interpretations can significantly influence how adaptation solutions are proposed and adopted (Becker et al., 2021; Cáceres et al., 2024). All other frameworks including adaptive capability theory, Cultural Theory, deductive theory, expected utility theory, existence theory, Marshallian demand, neoclassical production, protection motivation, random utility maximization (RUM), socio-cognitive, socio-ecological systems (SES), and sustainable intensification were referenced only once ($n = 1$ each), underscoring a lack of theoretical convergence.

Studies centered on adaptation strategies were heavily reliant on vulnerability framing, with isolated use of SES, adaptive capability, and sustainable intensification. RUM, previously assumed dominant, appeared only once. There was limited theoretical engagement ($n = 4$) for the awareness related studies with protection motivation and Marshallian demand each anchoring one study (Figure 8).

Half referenced theory implicitly, invoking institutional or political economy constraints. Perception related studies showed fragmented use of theory ($n = 8$), with vulnerability, deductive, cultural theory, and expected utility each appearing once. RUM again featured only once. Theoretical reliance skews toward economic rationality, with minimal integration of behavioral, cultural, or systems-based insights. This narrow framing constrains understanding of non-economic drivers—social norms, gendered barriers, power dynamics, and local knowledge—limiting cross-contextual synthesis. Embedding diverse theoretical lenses across adaptation, perception, and awareness domains is essential to strengthen conceptual coherence and foster actionable, context-sensitive adaptation research.

4.1.5 Target populations

In climate-change research on maize production among smallholders in low- and middle-income countries, there is a pronounced emphasis on smallholder maize farmers, who feature in 77% ($n = 20$) of studies. Smallholder farm households account for 12% ($n = 3$), smallholder coffee farmers for 8% ($n = 2$), and communal (subsistence) farmers for 4% ($n = 1$). Adaptation strategies emerged as the dominant theme, with 23 of 26 studies on on-farm innovations focusing exclusively on smallholder maize producers. These farmers are portrayed not as passive victims of climate stress but as active experimenters of context-specific solutions, though adaptive capacity varies with access to extension services, improved seed, irrigation, and institutional support.

Priority interventions consistently identified include drought-tolerant maize cultivars, water-conservation practices (e.g., zai pits, mulching), decentralized advisory systems, and weather-indexed insurance. Perception research, represented by five studies, highlights the acute vulnerability of smallholders cultivating less than two hectares under rain-fed conditions, with fieldwork in Eswatini, Ethiopia's Wolaita Zone, and Tanzania's Kongwa District documenting recurrent droughts, erratic rainfall, and heatwaves (Figure 9).

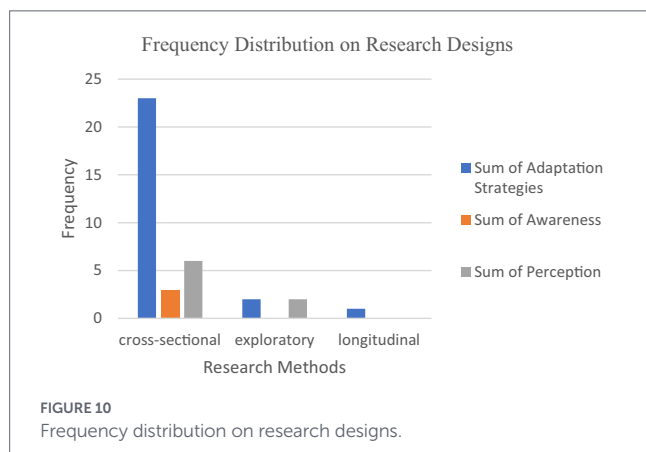
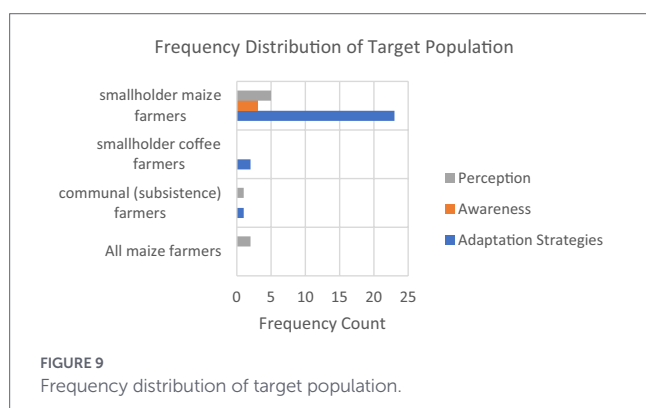
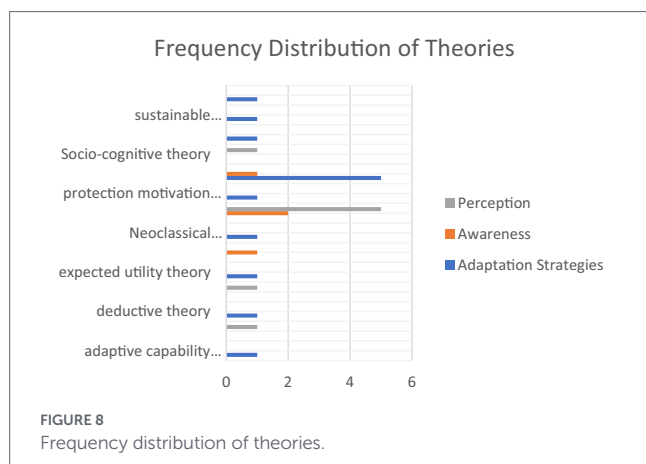
Broader analyses of all maize farmers (25%, $n = 2$) and mixed rural households (13%, $n = 1$) provide context but risk obscuring crop-specific adaptation needs. Awareness studies, largely labeling respondents as "smallholder farmers," reveal low awareness of climate risks and adaptation options in semi-arid Zimbabwean zones, with only one study explicitly defining its sample as "maize farmers."

This inconsistency in population taxonomy limits cross-study synthesis, while the concentration on maize-growing smallholders, though policy-relevant, narrows the research lens and thus future research could broaden target populations incorporating mixed cropping and agro-pastoral systems (e.g., coffee-maize intercropping), diverse household typologies, and livelihood-system frameworks that capture gender and equity dimensions. This diversification can uncover synergies across farming contexts, strengthen the evidence base on adaptive potential, and inform the design of nuanced, scalable interventions.

4.1.6 Research designs

Across sampled studies examining adaptation strategies, awareness, and perception of climate change among smallholder maize producers, research designs were overwhelmingly cross-sectional. Thirty-two studies (86%) employed cross-sectional surveys, four (11%) used exploratory approaches, and only one study (3%) adopted a longitudinal design. Under the adaptation strategies theme ($n = 26$), cross-sectional designs dominated (23 studies), with two exploratory and one longitudinal study. Perception-focused research ($n = 8$) also leaned heavily on cross-sectional methods (six studies) supplemented by two exploratory investigations. All three awareness-oriented studies were exclusively cross-sectional (Figure 10).

This reliance on one-off snapshots constrains the ability to infer causality, observe how farmers' adaptive behaviors evolve, or gauge the durability of interventions under shifting climatic stresses. Exploratory designs, though offering richer qualitative context, remain underutilized. The scarcity of longitudinal studies underscores a critical gap in capturing temporal dynamics. Common analytical tools across the cross-sectional and exploratory studies included multinomial logit



models to assess choice behaviors, multivariate probit frameworks for joint decision analysis, and classification and regression trees (CART) for non-parametric pattern discovery. However, without panel data or quasi-experimental controls, these methods can only suggest associations, not causal pathways.

To build more policy-relevant, nuanced evidence, future research should expand longitudinal panel surveys that track individual farm households across seasons and years. By integrating quasi-experimental or natural-experiment designs—such as difference-in-differences analyses around adaptation interventions—researchers can isolate treatment effects and establish causal mechanisms. Embedding robust theoretical frameworks within designed mixed-methods

sequences will further strengthen empirical rigor. Such methodological diversification will illuminate the trajectories and sustainability of smallholder adaptation, enabling policymakers to tailor climate-smart strategies that reflect both the lived realities and evolving needs of maize farming communities.

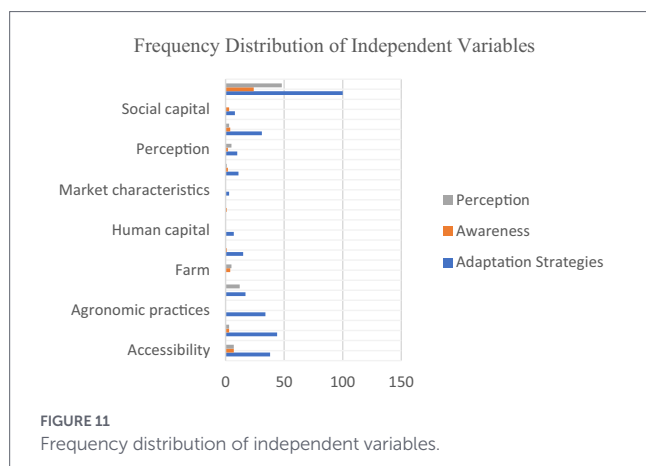
4.1.7 Independent variables

Across studies of climate change impacts among smallholder maize producers in LMICs, socioeconomic characteristics dominate the independent variables, accounting for 31% ($n = 100$) of measures. Adaptation strategies follow at 14% ($n = 44$), capturing on-farm adjustments such as crop diversification, conservation agriculture, and climate-resilient techniques, though their lower representation suggests insufficient depth in assessing strategy diversity and effectiveness. Accessibility variables including extension services, market and credit access, infrastructure, and climate information comprise 12% ($n = 38$), while agronomic practices, such as seed selection, fertilizer use, irrigation, and pest control, represent 11% ($n = 34$). Physical capital assets such as land ownership, livestock, and farm equipment make up 10% ($n = 31$).

Climate variables, surprisingly underrepresented, account for only 5% ($n = 17$), reflecting reliance on farmer perceptions over meteorological data. Financial capital also comprises 5% ($n = 15$). Membership in organizations, perception, and social capital each comprise 3% ($n = 11$; $n = 10$; $n = 8$, respectively), with human capital at 2% ($n = 7$) and market characteristics at 1% ($n = 3$) (Figure 11). Studies examining perception and awareness of climate change underscore that perceived risk and knowledge shape adaptive behavior, yet collectively these psychosocial dimensions remain underexplored. Overall, this distribution reveals a dominant focus on demographic and household determinants, with emerging but limited attention to behavioral, institutional, and technical drivers. Highlighting the themes of adaptation strategies, perception, and awareness, this synthesis underscores the need for integrative frameworks that incorporate market, infrastructural, and psychosocial variables to more comprehensively inform policy and support context-specific resilience-building among smallholder maize farmers. Expanding measures of awareness through educational outreach and climate literacy initiatives will contextualize the processes underlying adaptation decisions.

4.1.8 Mediating variables

Smallholder maize adaptation research overwhelmingly applies direct-effect models, linking determinants such as education, income, or land size directly to outcomes like the adoption of climate-smart practices. Only one study formally treated adaptation perception as a multidimensional mediating variable capturing exposure to climatic hazards, perceived sensitivity of maize systems, and perceived adaptive capacity to illustrate how cognition, motivation, and resource access jointly drive decisions. Established behavioral frameworks (Theory of Planned Behavior, Protection Motivation Theory, Risk Perception Models) suggest that perception, risk appraisal, self-efficacy, and intention mediate adaptation choices, yet none of the eight peer-reviewed studies reviewed formally applied mediation analysis. Similarly, within three articles focusing on awareness, zero included mediating constructs.



This gap obscures the nuanced pathways through which socio-cultural factors, institutional support, trust in information, and intra-household dynamics shape engagement in adaptation strategies. Neglecting mediators may lead policymakers to misjudge farmers' readiness for technical interventions or to overlook subjective barriers, such as perceiving climate change as an immediate threat. To advance both research and practice, integrating mediating variables via structural equation modeling, causal mediation methods, and mixed-methods designs is critical.

By unpacking how perception and awareness influence behavioral intentions alongside resource constraints, future studies can generate actionable insights. Policymakers and practitioners can then design gender-sensitive extension, peer learning, and risk communication campaigns that align with farmers' lived realities, promoting more equitable and context-appropriate adaptation strategies. One focus group study inferred mediating effects but lacked formal mediation frameworks. This underscores the need for mediation hypotheses.

4.1.9 Moderating variables

Across 26 studies of adaptation strategies, 3 of awareness, and 8 of perception among smallholder maize producers in LMICs, none incorporated moderating variables (100%, $n = 26$; 100%, $n = 3$; 100%, $n = 8$). This uniform omission implies an assumption of homogeneity in climate impacts and responses across farmer populations. Hints of subgroup differences appear only in qualitative accounts, which note gender and age disparities in access to land, information, or credit, yet lack rigorous theoretical framing or statistical interaction tests.

Reliance on descriptive and main-effect models limits understanding of why certain groups are more vulnerable or successful in adapting to climate stressors. Without moderation analysis, the literature cannot reveal how contextual factors—such as gender, education, extension services, social capital, or agro-ecological variation—alter the strength or direction of climate-adaptation relationships. This analytical gap obscures critical heterogeneity in adaptation capacity and effectiveness across diverse smallholder contexts.

Methodological advancement demands quantitative designs that include interaction terms and subgroup analyses to test whether the effect of climate variability on adaptation behaviors differs by socio-demographic attributes or resource access. Qualitative research should adopt comparative frameworks that explicitly explore subgroup contrasts and contextual conditioning factors. For policy and programming, this gap risks promoting one-size-fits-all solutions that may be ineffective or inequitable. Understanding moderation could reveal, for

example, that women-headed households or farmers with limited market access require tailored credit packages or differentiated institutional support mechanisms.

Incorporating moderating variables is essential to deepen theoretical models of differential vulnerability and resilience and to inform evidence-based, context-sensitive interventions. Future research integrating moderation analysis will enable more nuanced insights into awareness, perception, and adaptation strategies, thereby strengthening resilience among diverse smallholder maize producers in low- and middle-income countries and promoting equitable, locally responsive resilience-building outcomes.

5 Smallholder maize farmers climate change perception

5.1 Gaps in theory

Despite a growing body of literature on climate change and smallholder adaptation, the theoretical underpinnings remain fragmented and underdeveloped. Many studies rely on behavioral or economic decision-making models without embedding them within comprehensive, interdisciplinary frameworks. For instance, Adeagbo et al. (2021) utilize socio-ecological systems theory to assess adaptation strategies among Nigerian maize farmers, yet the framework does not account for long-term behavioral change or resilience-building. Similarly, Belay et al. (2017) emphasize socioeconomic drivers of adaptation but fall short of linking these to structural inequalities or historical marginalization that may influence adaptive capacity.

Moreover, studies often treat constructs like awareness and perception as static, self-reported variables, without integrating them into established psychological theories such as the Theory of Planned Behavior (Ajzen, 1991) or Protection Motivation Theory. For example, Boimah (2023) constructs awareness indices using binary indicators but does not engage with underlying psychological drivers of behavior. Martey et al. (2022) reference the “social amplification of risk” theory but do not apply it systematically across contexts. This weak theoretical integration limits understanding of how cognition, motivation, and social influence jointly shape adaptation behavior.

In addition, climate justice, vulnerability, and political economy perspectives remain marginal in adaptation research. While Mugisho et al. (2024) and Mukarumbwa (2022) document disparities in access to adaptation tools, these are not embedded within broader frameworks of vulnerability or environmental justice. Yet, different conceptualizations of vulnerability shape how adaptation priorities are defined and which groups are considered at risk (Becker et al., 2021). Gendered and communal dimensions of adaptation are also under-theorized. As noted by Kunene (2019), perception is often treated as an individual trait, neglecting its social and collective formation in rural communities.

Critically, none of the reviewed studies formally incorporated moderating variables such as gender, education, or institutional access, which obscures differential vulnerabilities and adaptation pathways among farmer groups. This theoretical omission limits the field's ability to explain why similar climate exposures produce divergent outcomes across households and regions. Greater integration of behavioral, socio-ecological, vulnerability, and climate justice frameworks would allow researchers to better capture the

multi-dimensional and unequal nature of adaptation processes (Becker et al., 2021). Future research should adopt integrated theoretical models that combine behavioral theories, socio-ecological systems thinking, vulnerability frameworks, and justice-oriented perspectives to explain heterogeneous adaptation pathways, structural barriers, and long-term resilience outcomes among smallholder maize producers.

5.2 Empirical gaps

Empirical studies on climate change impacts among smallholder maize farmers have generated valuable insights, but several gaps limit their depth, rigor, and policy relevance. A major issue is the heavy reliance on cross-sectional data, which captures behavior at a single point in time but fails to track adaptation across evolving climate conditions. Adeagbo et al. (2021), Belay et al. (2017), and Madamombe et al. (2023) all use cross-sectional survey designs, restricting their ability to assess long-term behavioral change or policy impacts.

Another critical gap is the limited triangulation with objective climate data. Most studies rely on farmers' self-reported perceptions of rainfall and temperature change, rarely validating these against meteorological records or productivity outcomes. For example, Teshome et al. (2021) compare perception data with climate trends but do not statistically test their alignment. Manzvera et al. (2024) examine willingness to pay for weather forecasts without assessing whether farmers' perceptions reflect actual climatic variability. This weakens empirical confidence in reported climate–adaptation linkages.

Geographic concentration further constrains generalizability. Research is heavily focused on Ghana, Nigeria, Ethiopia, and Zimbabwe, leaving fragile zones, peri-urban areas, and conflict-prone regions underexplored (Boimah, 2023; Martey et al., 2022). This spatial bias risks masking unique vulnerability profiles and adaptation constraints in underrepresented regions. Social and institutional mediators also remain poorly captured. Few studies examine how social networks, institutional trust, or indigenous knowledge systems shape the relationship between perception and adaptation. As a result, the pathways through which information, norms, and authority influence farmer behavior remain unclear.

Finally, a considerable number of studies exhibit limited analytical rigor. Madamombe et al. (2023) rely primarily on descriptive statistics, while Manzvera et al. (2024) apply contingent valuation methods without addressing selection bias or robustness checks. Sample sizes are often small, and survey biases such as recall or non-response bias are rarely discussed. Overall, empirical research must evolve toward longitudinal, multi-site, and mixed-method designs that integrate objective climate data, disaggregated social variables, and stronger statistical validation to generate more reliable and policy-relevant evidence.

5.3 Contextual gaps

Climate change adaptation research often overlooks the diverse contexts within which smallholder maize farmers operate. A persistent assumption of household homogeneity ignores variation based on gender, ethnicity, land tenure, and institutional access. Belay et al. (2017) include gender and household size as variables but do not disaggregate findings to assess how these shape adaptation decisions. Mugisho et al. (2024) document disparities in resource access but do not link them to broader socio-political structures.

Cultural, political, and historical contexts are similarly underexplored. Boimah (2023) examines environmental awareness in

Northern Ghana without considering how land policies, governance systems, or historical marginalization influence perception and response. Manzvera et al. (2024) acknowledge geographic disparities in Zimbabwe but do not analyze how institutional trust or historical exclusion affect farmers' investment in climate information.

Communication ecosystems also receive limited attention. Although radio ownership is frequently measured (e.g., Manzvera et al., 2024), few studies examine how climate messages are framed, which information sources are trusted, or how knowledge circulates within communities. The internal dynamics of social networks—including power relations, gender inclusion, and youth participation—remain largely unexplored (Kunene, 2019).

Indigenous knowledge systems are also weakly integrated. While Mbegalo et al. (2024) mention traditional forecasting practices, few studies examine how these interact with scientific climate information or influence adaptation decisions. This omission risks designing interventions that are culturally misaligned or poorly trusted by local communities.

Gendered dimensions of adaptation remain marginal. Although women's constraints in land, finance, and information access are widely recognized, their adaptation strategies and outcomes are rarely analyzed separately (Kunene, 2019). Intersectional factors such as age, class, and ethnicity are virtually absent. Contextual gaps limit understanding of how social identity, governance, culture, and information systems shape adaptation. Future research must incorporate disaggregated data, political–institutional analysis, and culturally grounded approaches to reflect the lived realities of diverse smallholder maize farmers.

5.4 Methodological gaps

Methodological weaknesses are evident across many studies on climate change adaptation among smallholder maize farmers. Quantitative approaches, particularly regression-based surveys, dominate the field (Boimah, 2023; Adeagbo et al., 2021). While effective for identifying statistical associations, these methods often overlook the subjective, cultural, and experiential dimensions of farmer decision-making.

Measurement validity is another concern. Many studies use binary or Likert-scale indicators to measure perception and awareness without psychometric validation. Boimah (2023) constructs awareness indices using yes/no responses, while Teshome et al. (2021) do not test reliability or construct validity. This limits confidence in how accurately key psychological constructs are captured.

Longitudinal and experimental designs are virtually absent. Most studies, including Adeagbo et al. (2021) and Madamombe et al. (2023), rely on one-off surveys, preventing assessment of whether adaptation strategies persist, evolve, or fail over time. Megersa et al. (2022) incorporate meteorological trend data but do not link it to household-level behavioral change.

Sampling procedures are often poorly described. Manzvera et al. (2024), for example, apply contingent valuation methods without addressing non-response bias or sample selection issues. Qualitative and participatory methods such as focus groups, interviews, and participatory mapping remain underutilized despite their value for capturing lived experience and indigenous knowledge.

Advanced tools such as GIS, remote sensing, and social network analysis are largely absent, limiting the ability to spatially map climate exposure or track information flows within communities (Isaac et al., 2009).

To strengthen methodological rigor, future studies should adopt mixed-methods designs, longitudinal panel data, causal inference techniques, validated measurement instruments, and spatial or network-based tools to better capture the dynamic, social, and environmental dimensions of climate adaptation.

6 Limitations

While this study offers important insights into how smallholder maize farmers adapt to climatic stresses, several methodological and contextual constraints must be acknowledged. These limitations may influence the strength of causal inferences, the breadth of applicability, and the completeness of the adaptation picture we present. Recognizing these gaps is crucial for interpreting our findings accurately and for guiding more robust future research.

- *Reliance on cross-sectional, self-reported data:* this approach precludes tracking adaptation trajectories over time and weakens causal claims. It also exposes findings to recall and social desirability biases, risking misrepresentation of farmers' behaviors and perceptions.
- *Underreporting of methodological constraints:* as only one-third to two-fifths of studies disclose their limitations, the existing literature demonstrates a notable lack of fundamental transparency. This opacity prevents rigorous appraisal, undermines meta-analyses, and can mislead policymakers about the reliability and scope of the evidence base.
- *Narrow geographic scope and low data granularity:* most research is limited to single districts or provinces and often relies on aggregated regional climate proxies. Such confined spatial coverage and coarse data reduce external validity and obscure fine-scale climate impacts and adaptation needs across diverse agro-ecological settings.
- *Exclusion of key variables and insufficient disaggregation:* omitting livelihood dimensions (such as livestock practices, off-farm income, multi-crop systems) and aggregating social or crop cohorts oversimplifies the adaptation landscape. This masks important heterogeneity in exposure, sensitivity, and strategy overlaps among different farmer groups.
- *Data scarcity and analytical shortcomings:* the absence of longitudinal panels, small sample sizes, potential endogeneity, and unvalidated survey tools all erode the robustness of statistical inferences. One-off datasets and single-method studies limit replicability and generalizability, weakening the foundation for sound adaptation policy.

7 Conclusion

This scoping review synthesized empirical evidence from 38 peer-reviewed studies published between 2008 and 2025 to assess the impact of climate change on smallholder maize producers in low- and middle-income countries. The findings demonstrate that climate change significantly disrupts smallholder maize farming, affecting yield reliability, household income, and food security. Farmers actively engage in adaptation strategies—including adjusting planting dates, adopting drought-resistant varieties, practicing intercropping, implementing conservation agriculture, and seeking off-farm income—highlighting their proactive role in mitigating climatic risks.

Adaptation strategies dominated the research landscape, accounting for 70.3% ($n = 26$) of studies, followed by perception (21.6%, $n = 8$) and awareness (8.1%, $n = 3$). This distribution reveals that while adaptation practices have been extensively documented, perception and awareness—which fundamentally shape adaptive behavior—remain underexplored. The narrow focus on production metrics often neglects broader socio-economic consequences such as household nutrition, income stability, and market access, indicating critical evidence gaps in food security outcomes and livelihood resilience.

Methodologically, cross-sectional designs dominated (86%, $n = 32$), with exploratory approaches (11%, $n = 4$) and longitudinal studies (3%, $n = 1$) being scarce. This reliance on one-time snapshots restricts understanding of adaptation effectiveness, behavioral trajectories, and intervention sustainability over time. Geographically, research concentrated heavily in Ethiopia (19%), Ghana (19%), and Zimbabwe (14%), leaving other regions and agro-ecological contexts underrepresented. Theoretical grounding varied considerably: 28% of studies lacked any framework, 20% employed random utility maximization theory, and the remaining 52% drew on 13 diverse theories—each accounting for only 4% of studies—limiting conceptual coherence and cross-study synthesis.

Critically, no study incorporated moderating variables to examine how gender, education, extension access, or agro-ecological context condition climate-adaptation relationships. Similarly, mediation analysis was virtually absent, obscuring the pathways through which socio-economic factors, institutional support, and farmer perceptions translate into adaptive action. These analytical gaps constrain understanding of differential vulnerability and limit the development of equitable, context-sensitive interventions.

Structural barriers—including gender inequality, limited extension services, poor institutional support, and insecure land tenure—compound climate risks and contribute to highly uneven adaptive capacity within and across communities. Current research often lacks rigorous frameworks, representative sampling, longitudinal designs, and dynamic methodologies needed to reflect real-world complexity and inform effective policy responses.

Going forward, climate adaptation research must evolve from narrow, technical assessments to holistic, interdisciplinary inquiry that centers equity, context, and transformation. Future studies should prioritize longitudinal and quasi-experimental designs, incorporate mediating and moderating variables, expand geographic and theoretical diversity, and explicitly integrate socio-economic, gendered, institutional, and policy dimensions. Only through such comprehensive approaches can smallholder maize farmers be effectively supported in building resilient and sustainable livelihoods under escalating climate stress.

8 Recommendation

Based on the findings and identified gaps, the following recommendations are proposed for research, policy, and practice:

8.1 Research recommendations

Drawing on the identified gaps, several research priorities emerge as essential for advancing understanding of climate change impacts on smallholder maize producers.

- First, future studies should adopt longitudinal and mixed-methods research designs to capture the temporal dynamics of adaptation—how farmer behaviors evolve in response to repeated climate shocks, policy changes, and learning processes. Combining survey data with qualitative insights from participatory rural appraisal or ethnographic fieldwork will generate context-sensitive findings that reflect the lived experiences of smallholder communities (Boimah, 2023; Teshome et al., 2021). Experimental and quasi-experimental designs, such as difference-in-differences analyses, are critical for establishing causal relationships and assessing the sustainability and effectiveness of interventions across multiple growing seasons.
- Second, research must expand beyond yield-focused metrics to incorporate comprehensive indicators of food security and livelihoods. This entails evaluating dimensions of availability, access, utilization, and stability, thereby capturing the broader socio-economic consequences of climate variability on maize-dependent households. Integrating multidimensional measures ensures that adaptation strategies are assessed not only for productivity gains but also for their capacity to sustain household well-being, income stability, and nutritional security.
- Third, studies should investigate the role of informal institutions, social networks, and indigenous knowledge systems in shaping perception and adaptive behavior. Farmers often rely on kinship ties, religious organizations, and local leadership rather than formal extension services for climate information and decision support. Understanding these informal systems is critical, particularly in settings where institutional trust is low or state capacity is limited (Belay et al., 2017; Mbegalo et al., 2024). Incorporating social network analysis and participatory methods can reveal how information flows and collective action influence adaptation outcomes.
- Fourth, theoretical frameworks should be consistently and rigorously applied to guide analysis and interpretation. Frameworks such as Resilience Theory, the Sustainable Livelihoods Framework, Vulnerability Analysis, and socio-ecological systems approaches can deepen understanding of adaptation processes, elucidate complex interactions among socio-economic and environmental variables, and enhance the policy relevance of findings. Theoretical coherence across studies will facilitate cross-contextual synthesis and strengthen the evidence base for scalable interventions.
- Fifth, research must increase geographic and agro-ecological diversity. Comparative studies across various climatic zones and regions can uncover context-specific adaptation strategies and identify practices that are scalable and effective under differing environmental conditions. Addressing the current concentration of research in Ethiopia, Ghana, and Zimbabwe will generate more representative evidence and support equitable policy development across Sub-Saharan Africa and other LMICs.
- Finally, future research must explicitly address gendered, socio-cultural, and intersectional dimensions of adaptation. This includes investigating how gender roles, land tenure systems, education levels, cultural practices, and ethnic identities influence adaptation uptake and outcomes. Disaggregated analyses are crucial to uncover the uneven distribution of climate risk and adaptive capacity among women, youth, and marginalized groups (Mugisho et al., 2024; Kunene, 2019). Incorporating moderating and mediating variables—such as access to credit,

extension services, and decision-making power—will reveal the pathways through which structural inequalities shape vulnerability and resilience. Ensuring that studies account for these dimensions promotes inclusive and equitable interventions that support all smallholder farmers in building climate-resilient livelihoods.

8.2 Policy recommendations

Several policy interventions emerge from the review's findings.

- Integration of climate services into extension systems is paramount. Providing localized weather forecasts, agronomic advice, and climate-smart information can significantly improve farmer awareness and the adoption of adaptation practices. Effective climate services bridge the gap between scientific knowledge and actionable decisions at the farm level.
- Investment in context-specific adaptation trials is also crucial. Demonstration plots, drought-tolerant maize varieties, water harvesting infrastructure, and conservation agriculture interventions should be promoted, evaluated, and scaled across diverse agro-ecological zones. Tailoring these interventions to improve local conditions ensures their effectiveness and encourages widespread adoption.
- Strengthening institutional and governance support is another priority. Policies should facilitate access to credit, insurance, agricultural inputs, and market linkages, thereby enhancing smallholder resilience. Multi-level governance approaches that align local adaptation actions with national and regional climate strategies can improve coordination and the overall effectiveness of adaptation efforts.
- Promotion of multi-dimensional resilience monitoring is equally important. Policy frameworks should incorporate indicators beyond yield, including food security, income stability, social capital, and environmental sustainability. Such holistic monitoring enables policymakers to assess the true impact of adaptation interventions and adjust strategies accordingly.
- Finally, support for interdisciplinary research and partnerships should be encouraged. Collaborative initiatives that integrate agronomy, socio-economics, nutrition, and climate science generate holistic insights, inform evidence-based policy decisions, and enhance the relevance and applicability of adaptation programs.

8.3 Practical recommendations for smallholder farmers

At the farm level, smallholder maize producers can implement several practical measures to enhance resilience.

- Adoption of diversified adaptation strategies is essential. Farmers should combine drought-tolerant maize varieties, intercropping, soil conservation, and water management practices to buffer against climatic variability and reduce risk exposure.
- Engagement with extension and climate advisory services is equally important. Leveraging available knowledge networks helps farmers make informed decisions, adopt innovative practices, and respond proactively to climate hazards.

- Enhancing post-harvest management and market access is another critical strategy. Reducing post-harvest losses and improving income stability complement on-farm adaptation efforts, ensuring that productivity gains translate into tangible benefits for households.
- Participation in farmer cooperatives and knowledge-sharing networks facilitates access to resources, innovation, and training. Collective action strengthens social capital, enhances bargaining power, and fosters a culture of shared learning, enabling smallholder farmers to implement adaptation strategies more effectively and sustainably.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary material](#), further inquiries can be directed to the corresponding author.

Author contributions

SG: Conceptualization, Data curation, Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing. DN: Conceptualization, Data curation, Methodology, Software, Supervision, Validation, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fclim.2026.1704769/full#supplementary-material>

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